

Answers

1. Compute the lengths of the following vectors:

a. $(2, 0)$ **Answer** $\sqrt{2^2 + 0^2}$
 $\sqrt{4}$
2

b. $(0, 3)$ **Answer** $\sqrt{0^2 + 3^2}$
 $\sqrt{9}$
3

c. $(3, 4)$ **Answer** $\sqrt{3^2 + 4^2}$
 $\sqrt{9 + 16}$
5

2. Compute the distance between the following pairs of points:

a. $(3, 4), (0, 0)$ **Answer** $\sqrt{(3 - 0)^2 + (4 - 0)^2}$
 $\sqrt{9 + 16}$
5

b. $(4, -5), (-4, 5)$ **Answer** $\sqrt{(4 - (-4))^2 + ((-5) - 5)^2}$
 $\sqrt{8^2 + (-10)^2}$
 $\sqrt{64 + 100}$
 $\sqrt{164}$
~12.81

3. Compute the length of the following 3D vector:

a. $(3, 4, 12)$ **Answer** $\sqrt{3^2 + 4^2 + 12^2}$
 $\sqrt{9 + 16 + 144}$
 $\sqrt{169}$
13